MILITARY SPECIFICATION

TEST SET, FIRE CONTROL SUBSYSTEM: M143

This specification is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains the performance, Quality Assurance Provisions, and test requirements for the Head-Up Display Subsystem (HUDS) Test Set to be used for the AH-1S Helicopter Aviation Intermediate Maintenance (AVIM). This Test Set is identified as Peculiar Ground Support Equipment (PGSE).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materials, General Specification Governing the Manufacture and Inspection of
MIL-T-28800	-	Test Equipment for use with Electronic and Electrical Equipment
MIL-M-38510	-	Microcircuits, General Specification for
MIL-S-45743	-	Soldering, Manual Type, High Reliability Electrical, Electronic for Aerospace and Control Systems, Procedure for
MIL-S-46844	-	Solder Bath, Soldering and Printed Wiring Assemblies, Automatic Machine Type

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-415	- Test Provisions for Electronic Systems and Associated Equipment, Design and Criteria for
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests
MIL-STD-883	- Test Methods and Procedures for Microelectronics
MIL-STD-1366	- Packaging, Handling, Storage, and Transportational System Dimensional and Weight Constraints, Definition of
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-45662	- Calibration Systems Requirements

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

11780001	- Display Unit, Sight, Head Up Subsystem: M76 Assembly
11780002	- Processor, Signal, Sight, Helicopter Head Up Subsystem, M76
11780600	- Test Set, Fire Control Subsystem: M143
11780704	- Holder Assembly, Electrical Card
11780665	- Chassis, Power Supply, Assembly
11780680	- Front Panel Assembly

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11780651 - Cable Assembly W1, Special Purpose, Electrical
11780654 - Cable Assembly W2, Special Purpose, Electrical
11780656 - Cable Assembly, Branched (W3) Special Purpose, Electrical
11780657 - Cable Assembly, W4, Special Purpose, Electrical
11780658 - Cable Assembly, W5, Special Purpose, Electrical
11780647 - Circuit Card Assembly Extender Board Test Set

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951 - Standard Practice for Commercial Packing

TECHNICAL MANUAL

TM-4931-379-13&P

(Applications for copies of ASTM publications should be addressed to the ASTM, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item definition. The Test Set shall meet the requirements of MIL-T-28800, as detailed herein. The Test Set shall provide all the necessary stimuli and measuring interface capability to functionally test the Sight, Helicopter, Display Subsystem: M76 (HUDS). The Test Set shall provide the capability for:

a. Testing the entire HUDS, consisting of a Signal Processor (SP) 11780002, and Head-Up Display (HUD), 11780001, as a system.

b. Testing the Signal Processor alone.

c. Testing the Head-Up Display alone.

Utilizing the Test Set, it shall be possible to fault isolate to a defective Shop Replaceable Unit (SRU). Allow 5 minutes "warm-up" time before normal test conditions.

3.1.1.2 Electrical interface definition. Test Set input/output signal and power listing is shown in Tables I through VI, allocated as follows:

Table I	Test Set Power Inputs
Table II	Signal Processor/Test Set Interface
Table III	SP/HUD/Test Set Interface
Table IV	Test Set Passive Inputs from Signal Processor (Test Points)
Table V	Test Set Passive Inputs from HUD (Test Points)
Table VI	Test Set outputs to Common Ground Support Equipment (CGSE)

3.1.1.3 Physical interface definition. The Test Set interconnections with the HUDS shall be via five cable assemblies:

a. Cable W1 (Drawing 11780651) providing Test Set Input power, twelve feet long.

b. Cable W2 (Drawing 11780654) providing Test Set output power and signals to the signal processor, six feet long.

c. Cable W3 (Drawing 11780656) providing SP/HUD/test Set, SP/Test Set, or HUD/Test Set interconnections, six feet long.

d. Cable W4 (Drawing 11780657) providing access to the SP test point connector, six feet long.

e. Cable W5 (Drawing 11780658) providing access to the HUD test point connector, six feet long.

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3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Capability. The Test Set shall be capable of testing the HUDS System to provide rapid fault isolation in support of AVIM level of maintenance.

3.2.1.1.1 Functions. The Test Set shall be capable of testing the HUDS as a system as well as each LRU individually. It shall provide all the necessary inputs to the HUDS and provide all the necessary output monitoring provisions for the performance of the following HUDS functional tests:

a. System acceptance testing

b. LRU acceptance testing

c. Fault isolation to the following defective SRU's:

Signal Processor LRU

A2	Low Voltage Power Supply Assembly
A3	Input Circuit Card Assembly
A4	CPU Circuit Card Assembly
A5	Output Circuit Card Assembly

Head-Up Display LRU

A1	CRT Assembly
A2	Bright Up Amplifier Assembly
A3	X - Deflection Assembly
A4	Y - Deflection Assembly
PS1	High Voltage Power Supply

3.2.1.1.2 Modes of operation. In providing the HUDS inputs, the Test Set shall provide the following modes of operation:

a. HUDS or SP Only Modes

AUTO - Provide test routines which exercise the capabilities of the SP and HUD and produce predetermined fixed or movable symbols on the HUD or monitor display, requiring a minimal amount of operator interaction.

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b. HUDS, SP Only, or HUD Only Modes

MANUAL - Provides the operator with capability of fault isolation to a defective SRU by the selection of specific input data and monitoring of selected output information.

3.2.1.1.3 Protection. The Test Set shall provide fail-safe and current-limiting circuitry to protect the HUDS from damage resulting from Test Set failure.

3.2.1.1.4 Time meter. The Test Set shall incorporate a time totaling meter.

3.2.1.2 System capability.

3.2.1.2.1 Power requirements. The Test Set, when used in conjunction with the HUDS, shall require 28 VDC ± 2 VDC at 40 amps maximum. Input power connections shall be provided via cable assembly W1.

3.2.1.5 Controls and indicators. The Test Set front panel controls and indicators shall meet the intent of MIL-T-28800 in accordance with the applicable sections of MIL-STD-454 and MIL-STD-1472.

3.2.2 Physical characteristics.

3.2.2.1 Weight. The overall weight of the Test Set shall not exceed 125 pounds.

3.2.2.2 Transportability. The Test Set shall be movable by military or commercial carriers over railways, highways, oceans, and airways in accordance with the requirements of MIL-STD-1366. After being moved by any of these modes, or combination of these modes, the equipment shall be operational without recalibration or maintenance.

3.2.2.3 Keying. Modules and connectors shall be keyed to prevent the insertion of a connector or module into an improper location within or on the Test Set. All cable assemblies shall be keyed to prevent improper connection to the Test Set or LRU's.

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3.2.3 Reliability. Reliability shall be in accordance with requirement 35 of MIL-STD-454 and as specified herein. The specified Mean Time Between Failure (MTBF) for continuous and intermittent operation, in accordance with MIL-STD-781, shall be 1000 hours under the environmental conditions specified in 3.2.5 herein.

3.2.4 Test provisions. Test points shall be in accordance with MIL-STD-415. Fault isolation of the test Set to the Tester Replaceable Units (TRU) shall be provided. Substitution of known good TRU's or probing through conformal coating as a means of fault isolation shall not be allowed.

3.2.4.1 Fault isolation to the TRU. Fault isolation to the defective TRU is accomplished by a combination of test points on each TRU and the Front Panel in accordance with MIL-STD-415.

3.2.4.2 Front panel test points. Front Panel Test Points are designed to meet the intent of MIL-STD-415.

a.	Less than 120 Vac/Spare	White
b.	Less than 28 Vdc - positive	Red
c.	28 Vdc or greater - positive	Red encircled with red
d.	Chassis ground	Black
e.	Referenced ground other than chassis ground	Grey
f.	Less than 28 Vdc - negative	Violet
g.	28 Vdc or greater - negative	Violet encircled with red
h.	Digital signal	Blue
i.	Analog signal	Yellow

Note: Encirclement by Red denotes danger.

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3.2.4.3 Quantitative requirements.

3.2.4.4 Preventative maintenance. The Test Set shall not require calibration in intervals of less than 1000 hours of operation as noted on the time totaling meter.

3.2.5 Environmental conditions (first article only). The Test Set shall meet the performance requirements of this specification during or after exposure to the environmental conditions specified herein, as applicable.

3.2.5.1 Temperature and humidity. The temperature and humidity limits shall be as follows.

3.2.5.1.1 Operating limits

Low Temperature:	-15°C
High Temperature:	+55°C
Relative Humidity:	95%, +5%, -0%

3.2.5.1.2 Non-operating limits

Low Temperature:	-62°C
High Temperature:	+85°C
Relative Humidity:	95%, +5%, -0%

3.2.5.2 Altitude limits

Operating:	15,000 ft.
Non-operating:	50,000 ft.

3.2.5.3 Vibration. The Test Set shall operate satisfactorily after being subjected to vibrations ranging from 5 to 55 Hz with a maximum g level equal to 3g's and a minimum time for vibration/axis equal to 45 minutes.

3.2.5.4 Shock, mechanical.

3.2.5.4.1 Pulse. The Test Set shall operate satisfactorily after being subjected to the vibration environment specified in section 3.9.5.1 of MIL-T-28800.

3.2.5.4.2 Transit drop. The Test Set shall operate satisfactorily after being subjected to a drop height of 12 inches for a maximum of 26 drops as specified in section 3.9.5.2 of MIL-T-28800.

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3.2.5.4.3 Bench handling. The Test Set shall operate satisfactorily after being subjected to the bench handling test specified in 3.9.5.3 of MIL-T-28800.

3.2.5.5 Drip proof. The Test Set shall operate satisfactorily after being subjected to the water resistance test specified in 3.9.6.3 of MIL-T-28800.

3.2.5.6 Salt atmosphere. The Test Set shall operate satisfactorily after being subjected to the salt atmosphere tests specified in 3.9.8.2 of MIL-T-28800 except as follows:

a. The salt solution shall be 5 percent.

b. The exposure time shall be 200 hours.

3.2.5.7 Acoustical noise. The Test Set shall attenuate self-generated noise of items such as fans, motors, and generators. The noise generated by the Test Set shall not exceed the values specified when measured in accordance with 4.5.7.2 of MIL-T-28800.

3.2.5.8 Fungus inert material. The Test Set shall be inherently fungus inert per MIL-T-28800 (paragraph 3.9.7).

3.2.7 Electromagnetic radiation. The Test Set shall meet the requirements of MIL-STD-461 equipment when tested to the following requirements: CE03, CS01, CS02, CS06, RE02, RS02, and RS03. When performing such tests, the Test Set shall be connected to various loads representing the units normally under test, as applicable.

3.3 Construction.

3.3.1 Material, processes, and parts. Material, process, and part selection shall be per 3.3 of MIL-T-28800. Microcircuits shall conform wherever possible, to the requirements of MIL-M-38510, Class B devices; should such devices be unobtainable, parts screened to the requirements of MIL-STD-883, Method 5004, Level B, shall be considered acceptable.

3.3.2 Interchangeability. The Test Set shall meet the interchangeability requirements described in MIL-STD-454, Requirement 7.

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3.3.3 Safety. Safety shall be in accordance with Requirement 1 of MIL-STD-454, in addition to (a) and (b):

a. Grounding. When radio interference filters are used in the power input, filters shall contain only capacitors between each side of the line and chassis which are sufficiently low in value to allow less than 5 milliamperes (mA) of current to flow to the chassis under the most adverse conditions of maximum power frequency and maximum voltage permitted by the individual test equipment specification. (The primary line shall be protected or fuzed).

b. Grounding of interconnecting equipment. All cables, input power, and interconnecting cables to other equipment being tested (with the exception of radio frequency cables), shall be provided with a grounding wire. The grounding wire shall be connected through a terminal or lug in the connector to the chassis or frame and shall be solely for the purpose of providing a ground potential. Power return wires are not to be used as grounding wires.

3.3.4 Preconditioning (burn-in). Each Test Set procured under this specification shall be subjected to a burn-in test as defined in Figure 1.

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TYPICAL TEMPERATURE CYCLE-ELECTRONIC AND AVIONIC EQUIPMENT

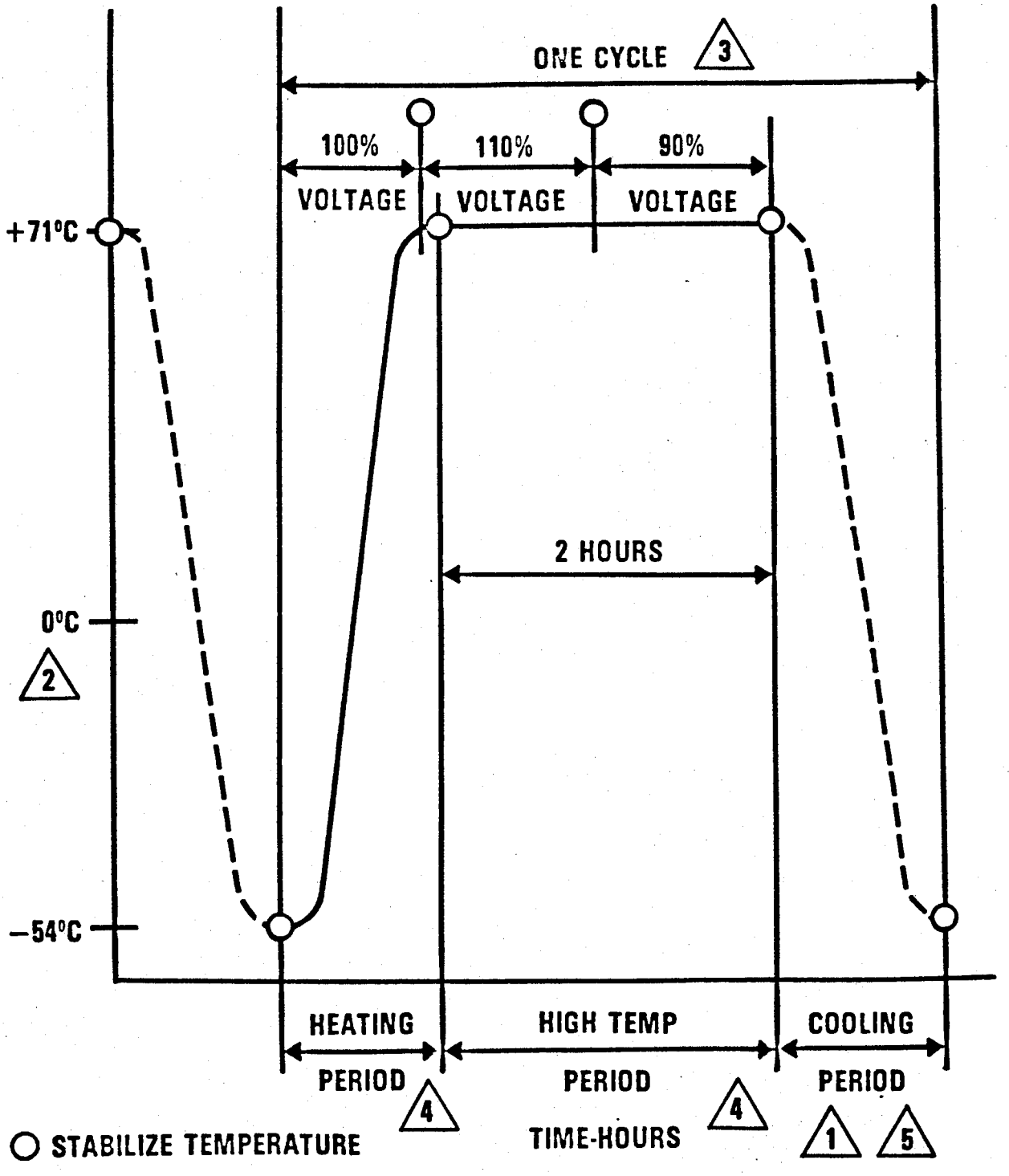


FIGURE 1. Typical temperature cycle - electronic and avionic equipment

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NOTES:

1. During the cooling period, the chamber temperature shall not be lower than -60°C.

2. Once every 24 hours at 0°C, moisture shall be injected until the equipment being tested is coated with frost.

3. One temperature cycle shall start with the equipment stabilized at the low temperature, heated and stabilized at the high temperature, maintained at the high temperature for 2 hours, then end with the equipment cooled and stabilized at the low temperature. During the temperature cycle, the equipment shall be operated in accordance with its normal duty cycle.

4. Equipment ON; input voltage at 110% and 90%, respectively during each 1/3 of the total time of heating period plus high temperature period; momentarily OFF between input voltage changes.

5. Equipment OFF.

6. Vibration -2.2g ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The vibration shall be at least 10 minutes during each hour of operating time.

7. Transition rate between temperature levels shall average 5°C per minute.

FIGURE 1. Typical temperature cycle - electronic and avionic equipment- Continued

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3.4 Major component characteristics.

3.4.1 Test set assembly. The Test Set assembly shall consist of the following:

a. Enclosure (drawing 11780600). A combination case with detachable cover. The case is an integral part of the Test Set, whereas the cover houses the cables and extender board described in 3.4.2 and 3.4.3.

b. Front Panel Subassembly (Drawing 11780680). Contains controls, switches, displays, etc., for providing the human interface to the Test Set for HUDS testing. The Front Panel is hinged to provide access to subassemblies within the case.

c. Power Cage Subassembly (Drawing 11780704). Mounted within the case and hidden by the Front Panel, this subassembly contains circuit card assemblies (referenced in Drawing 11780600) which provide all the digital and analog signals required by the HUDS.

3.4.2 Cable sets. The cable sets shall provide interconnections between the test Set assembly and HUDS. The cable sets are described in 3.1.2.2 of this document.

3.4.3 Extender board. The Test Set shall contain a universal extender board assembly (Drawing 11780649) to provide access to the components and interface of the circuit cards mounted within the Card Cage subassembly described in 3.4.1(d) of this document.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article by the contractor shall be as specified in the contract.

3.6 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

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Table I - Test Set Power Inputs

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J1-A	+28 VDC	DC	28 VDC	± 2 VDC	
J1-B	28 VDC RTN	DC Return	Ground		
J1-C	Ground				

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Table II - Signal Processor/Test Set Interface

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J2-A	115VAC 400Hz	AC Power	115VAC	±10VRMS	Single Phase
J2-W	115VAC 400Hz Return	AC Power Return	115VAC	400 Hz ± 20	
J2-D	Fixed Forward Gun	DC	True 28VDC False Open	±4VDC	
J2-F	Signal Return	DC	0VDC		Digital Ground
J2-G	HUD Data	TTL	True 5VDC False 0VDC	2.8VDC to 5VDC -0.5VDC to 0.8VDC	Serial data in binary 2's complement, 16 byte block transfer from differential linedriver.
J2-d	HUD Data Not	TTL Return			
J2-H	HUD Data Shield	Ground	0VDC		Digital Ground
J2-K	Raw LRF Data	TTL	True 5VDC False 0VDC	2.8VDC to 5VDC -0.5VDC to 0.8VDC	Serial data in binary 2's complement, 4 byte block transfer from differential linedriver.
J2-J	Raw LRF Data Not	TTL Return			Digital Ground
J2-e	Raw LRF Data Shield	Ground			Digital Ground

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Table II - Signal Processor/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J2-f	Az Coarse Pointer	Analog	±2.2V Max	±50MV	
J2-M	Az Coarse Pointer Return	Analog Ground			
J2-I	Az Coarse Pointer Shield	Chassis Ground			
J2-N	TCC Good	DC	True 28VDC False Open	±4VDC	
J2-P	HUDS Go/No Go	TTL Input	5VDC	±0.5VDC	No-Go=Low 0 VDC ±0.5V From HUDS
J2-R	Spare Discrete 2	DC	True 28VDC False Open	±4VDC 100K Ohms	
J2-T	+28VDC	DC Power	28VDC	±4VDC	
J2-Y	High Altitude Limit	DC	True 0VDC False Open	-0.5VDC +0.8VDC 100K Ohms	
J2-Z	Low Altitude Limit	DC	True 0VDC False Open	-0.5VDC +0.8VDC 100K Ohms	
J2-b	TSU EL Angle	Analog	±2.2ma Max	±0.05ma	Current Source
J2-c	TSU EL Angle	Analog Return			
J2-t	TSU EL Angle Shield	Ground			

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Table II - Signal Processor/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J2-g	Attack Mode	DC	True 28 VDC False Open	±4 VDC 100K Ohms	
J2-h	Spare	No Connection			
J2-i	Direct/Indirect	TTL Input	5 VDC		From HUDS Indirect=28VDC Direct=Open =100K Ohms
J2-x	Spare	No Connection			
J2-j	Descend Command	DC	True 10 VDC False Open	±1 VDC 100K Ohms	
J2-k	Ascend Command	DC	True 10 VDC False Open	±1 VDC 100K Ohms	
J2-q	Ready Mode	DC	True 28 VDC False Open	±4 VDC 100K Ohms	
J2-v	TSU Az Angle	Analog	±2.2 ma Max	±0.05 ma	Current Source
J2-w	TSU Az Angle Return	Analog Return			
J2-u	TSU Az Angle Shield	Ground			
J2-EE	Right Missile Select	DC	True 0 VDC False Open	-0.5 VDC +0.08 VDC 100K Ohms	

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Table II - Signal Processor/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J2-FF	Left Missile Select	DC	True 0 VDC False Open	-0.5 VDC +0.08 VDC 100K Ohms	
J2-B	Spare	No Connection			
J2-C	Spare Discrete 1	DC	True 28 VDC False Open	±4 VDC 100K Ohms	Not used at this time.
J2-E	Chassis Ground				
J2-S	Spare	No Connection			
J2-V	Spare	No Connection			
J2-a	Spare	No Connection			
J2-n	Spare	No Connection			
J2-p	Spare	No Connection			
J2-s	Spare	No Connection			
J2-y	Spare	No Connection			
J2-z	Spare	No Connection			
J2-AA	Spare	No Connection			
J2-BB	Spare	No Connection			
J2-CC	Spare	No Connection			
J2-DD	Spare	No Connection			
J2-GG	Spare	No Connection			
J2-HH	Spare	No Connection			
J2-U	+28 VDC	DC Power	28 VDC	±4 VDC	
J2-r	+28 VDC Return	DC Power Return	0 VDC		
J2-m	+28 VDC Return	DC Power Return	0 VDC		
J2-X	Fire Maneuver Mode	DC	True 28 VDC False Open	±4 VDC	

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Table III - SP/HUD/Test Set Interface

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J3/J4-b	HVPS Power	DC	J3 10 Ohms J4 +26 VDC	± 1 Ohms ± 2.6 VDC	J3 10 Ohms Load J4 Power Supply
J3/J4-c	HVPS Power Return	DC Return			
J3/J4-A	HVPS Power Shield	Ground			
J3/J4-B	Filament Power	J3-RES J4-DC	J3 12 Ohms J4 +9.0 VDC	± 1 Ohms ± 0.9 VDC	J3 12 Ohm Load J4 Power Supply
J3/J4-d	Filament Power Return	DC Return			J3 Load Return J4 Power Supply Return
J3/J4-C	Filament Shield	Ground			
J3/J4-D	+X/Y Deflection Signal Power	DC	J3 40 Ohms J4 +16 VDC	± 2 Ohms ± 1.6 VDC	J3 40 Ohm Load J4 Power Supply
J3/J4-E	+X/Y Deflection Signal Power Return	DC Return			J3 Load Return J4 Power Supply Return
J3/J4-f	-X/Y Deflection Signal Power	DC	J3 40 Ohms J4 -16 VDC	± 2 Ohms ± 1.6 VDC	J3 40 Ohm Load J4 Power Supply
J3/J4-e	±X/Y Deflection Signal Power Shield	Ground			

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Table III - SP/HUD/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J3/J4-G	Chassis Ground	Ground			
J3/J4-J	+X	Analog	± 5 VDC	MAX	J3 From SP J4 to HUD
J3/J4-K	-X	Analog	± 5 VDC	MAX	J3 From SP J4 to HUD
J3/J4-i	±X Shield	Ground			
J3/J4-P	-Y	Analog	± 5 VDC	MAX	J3 From SP J4 to HUD
J3/J4-R	+Y	Analog	± 5 VDC	MAX	J3 From SP J4 to HUD
J3/J4-n	±Y Shield	Ground			
J3/J4-X	±Z Amp Power Return	DC Return			J3 Load Return J4 Power Supply Return
J3/J4-Y	-Z Amp Power	DC	J3 3K Ohms J4 -85 VDC	± 30 Ohms	J3 3K Ohms Load J4 Power Supply
J3/J4-u	+Z Amp Power	DC	J3 2.2K Ohms J4 +85 VDC	± 22.0 Ohms ± 8.5 VDC	J3 2.2K Ohms Load J4 Power Supply
J3/J4-t	±Z Amp Power Shield	Ground			

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Table III - SP/HUD/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J3/J4-Z	115 VAC	AC			J3 Power from SP-No Connection J4 Power to HUD
J3/J4-v	115 VAC Return	AC Return	115 VAC		J3 Power from SP-No Connection J4 Power to HUD
J3/J4-a	115 VAC Shield	Ground			
J3/J4-g	Signal Return	DC Return			J3 to switch in Test Set-Ground Supply from SP J4 to Digital Ground
J3/J4-h	SP Fail	DC	True 5 VDC False 0 VDC	2.8 VDC to 5.5 VDC ± 0.5 VDC	J3 from SP to SYM Monitor LED J4 TIED TO Dig. Gnd. (Always False)
J3/J4-y	HUD Fail		True 5 VDC False 0 VDC	2.8 VDC to 5.5 VDC	J3 TIED to Dig. Gnd. (Always False) J4 from HUD to SYM Monitor LED
J3/J4-x	Signal Shield	Ground			
J3/J4-j	Bright-Up	TTL	0 - +5 VDC	± 0.5 VDC	J3 to Monitor (2) Differential J4 to HUD

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Table III - SP/HUD/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J3/J4-m	Bright-Up Not	TTL	0 - +5 VDC	2.8 VDC to 5.5 VDC	J3 to Monitor (2) Digital Sig. J4 to HUD
J3/J4-k	Bright-Up Shield	Ground			
J3/J4-p	Stadiametric Mode	DC	True 0 VDC False Open	+ 0.8 VDC	J3 Switch in Test Set-Active Low J4 Switch on HUD-Active Low
J3/J4-q	Test Mode	DC	True 0 VDC False Open	Open 100K Ohms	J3 Switch in Test Set-Active Low J4 Switch in Test Set-Active Low
J3/J4-DD	Stadiametric/ Test Mode Shield	Ground			
J3/J4-AA	HUDS Return	DC Return	J3-Ground when connected to SP		J3 to HUDS PWR SEL Switch J4 to Digital Gnd.
J3/J4-CC	HUDS On	DC	True 0 VDC False Open	+ 0.8 VDC	J3 Switch in Test Set-Active Low J4 Switch on HUD-Active Low
J3/J4-LL	HUDS Standby	DC	True 0 VDC False Open	Open 100K Ohms	J3 Switch in Test Set-Active Low J4 Switch on HUD-Active Low

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Table III - SP/HUD/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J3/J4-BB	HUDS Shield	Ground			
J3/J4-FF	-X/Y Deflection Booster Power	DC	J3 200 Ohms J4 -40 VDC	± 10 Ohms ± 4 VDC	J3 200 Ohms Load J4 Power Supply
J3/J4-HH	+X/Y Deflection Booster Power	DC	J3 200 Ohms J4 40 VDC	± 10 Ohms ± 4 VDC	J3 200 Ohm Load J4 Power Supply
J3/J4-NN	±X/Y Deflection Booster Power Return	DC Return			J3 Load Return J4 Power Supply Return
J3/J4-GG	±X/Y Deflection Booster Power Shield	Ground			
J3/J4-KK	Direct/Indirect	DC J4 Only	Indirect 28 VDC Direct Open	± 4 VDC J3 Open	J3 Switch in Test Set J4 Switch on HUD
J3/J4-MM	+ 28 VDC	DC	J3 = Open J4 = 28 VDC	± 4 VDC	J3 28 VDC to Direct/ Indirect Switch in T.S. from SP J4 28 VDC from T.S. to HUD
J3/J4-JJ	Shield	Ground			Direct/Indirect Switch

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Table III - SP/HUD/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J3/J4-F	Spare	No Connection			
J3/J4-H	Spare	No Connection			
J3/J4-L	Spare	No Connection			
J3/J4-M	Spare	No Connection			
J3/J4-N	Spare	No Connection			
J3/J4-S	Spare	No Connection			
J3/J4-T	Spare	No Connection			
J3/J4-U	Spare	No Connection			
J3/J4-V	Spare	No Connection			
J3/J4-r	Spare	No Connection			
J3/J4-w	Spare	No Connection			
J3/J4-z	Spare	No Connection			
J3/J4-pp	Spare	No Connection			
J3/J4-W	Spare	No Connection			
J3/J4-S	Spare	No Connection			
J3-EE	Spare	No Connection			
J4-W	Cockpit Lighting	On 28 VDC Off Open	± 4 VDC		
J7-A	Cockpit Lighting	On 28 VDC Off Open	± 4 VDC		
J4-s	Cockpit Lighting Return	28 V Return Ground			Ground
J7-B	Cockpit Lighting Return	28 V Return Ground			Ground

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Table III - SP/HUD/Test Set Interface (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J4-EE	Displays	On 28 VDC	± 4 VDV		Return is through 28 V Return
J7-C	Displays	On 28 VDC	± 4 VDV		Return is through 28 V Return
J7-D	Spare	No Connection			
J7-E	Spare	No Connection			

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Table IV - SP Test Points

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J5-Y	$\overline{A11}$	TTL	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-Y
J5-Z	$\overline{A12}$	TTL	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-Z
J5-a	$\overline{A13}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-a
J5-c	$\overline{A14}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-b
J5-c	$\overline{A15}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-c
J5-d	$\overline{D0}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-d
J5-e	$\overline{D1}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-e
J5-f	$\overline{D2}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-f
J5-g	$\overline{D3}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-g
J5-h	$\overline{D4}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-h
J5-i	$\overline{D5}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-i
J5-j	$\overline{D6}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-j
J5-k	$\overline{D7}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-k
J5-m	Sync	TTL Input	3.3 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-m
J5-n	$\overline{Reset}$	TTL Output	Not Used	At This Time	To Test Point TP 9-n

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Table IV - SP Test Points (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J5-p	$\overline{\text{Ready}}$	TTL Output	High 3.3 VDC Low 0 VDC	2.8 VDC to 5.5 VDC -0.5 VDC to 0.8 VDC	To Test Point TP 9-p
J5-q	$\overline{\text{B/U Ovrd}}$	TTL Output	High 3.3 VDC Low 0 VDC	2.8 VDC to 5.5 VDC -0.5 VDC to 0.8 VDC	To Test Point TP 9-q
J5-r	$\overline{\text{Test Fail}}$	TTL Output	On 0 VDC Off Open	-0.5 VDC to 0.8 VDC Momentary Switch	To Test Point TP 9-r
J5-s	$\overline{\text{Sym Parity}}$	TTL Input	Open		To Test Point TP 9-s Only
J5-t	Proc TP1	TTL Input	Open		To Test Point TP 9-t Only
J5-u	Proc TP2	TTL Input	Open		To Test Point TP 9-u Only
J5-v	$\overline{\text{Ack IN 6}}$	TTL Input	Open		To Test Point TO 9-v Only
J5-w	$\overline{\text{Frame Reset Override}}$	TTL Output	High 3.3 VDC Low 0 VDC	2.8 to 5.5 VDC -0.5 to +0.8 VDC	To Test Point TP 9-w
J5-x	$\overline{\text{Int 6}}$	TTL Input	Open		To Test Point TP 9-x Only
J5-y	F.R. Reset	TTL Input	5 VDC	2.8 V	To Test Point TP 9-y
J5-z	Sys Clk	TTL Input	5 VDC		To Test Point TP 9-z
J5-AA	TM Wd	TTL Input	Open		To Test Point TP 9-AA Only
J5-BB	$\overline{\text{Memory R/W}}$	TTL Input	5 VDC	2.8 VDC to 5.5VDC	To Test Point TP 9-BB

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Table IV - SP Test Points (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J5-CC	$\overline{\text{CPU Override}}$	TTL Output	High 3.3 VDC Low 0 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-CC
J5-DD	Input Test	TTL Output	Open		To Test Point TP 9-DD
J5-EE	$\overline{\text{UART Override}}$	TTL Output			To Test Point TP 9-EE
J5-KK	$\overline{\text{AO}}$	TTL Input	5 VDC	2.8 VDC to 5.5 VDC	To Test Point TP 9-KK
J5-G	Chassis Gnd	DC, AC Ground			To Test Point TP 9-G
J5-FF	Spare	No Connection			To Test Point TP 9-FF
J5-GG	Spare	No Connection			To Test Point TP 9-GG
J5-HH	Spare	No Connection			To Test Point TP 9-HH
J5-JJ	Spare	No Connection			To Test Point TP 9-JJ
J5-NN	Spare	No Connection			To Test Point TP 9-NN
J5-LL	Spare	No Connection			To Test Point TP 9-LL
J5-MM	Spare	No Connection			To Test Point TP 9-MM
J5-PP	Spare	No Connection			To Test Point TP 9-PP

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Table V - HUD Test Points

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J6-A	+12 VDC from Flex Print	Passive Test Point	Open		Test Point TP 10-A
J6-B	-12 VDC from Flex Print	Passive Test Point	Open		Test Point TP 10-B
J6-C	Grid 1 80V P-P	Passive Test Point	Open		Test Point TP 10-C
J6-E	X Yoke Return	Passive Test Point	Open		Test Point TP 10-E
J6-F	X Input Slow	Passive Test Point	Open		Test Point TP 10-F
J6-G	X Sweep Slow	Passive Test Point	Open		Test Point TP 10-G
J6-H		Passive Test Point	Open		Test Point TP 10-H
J6-J	Cathode 60V P-P	Passive Test Point	Open		Test Point TP 10-J
J6-K	HVPS On/Off	Passive Test Point	Open		Test Point TP 10-K
J6-L	HUD BIT indicator	Passive Test Point	Open		Test Point TP 10-L
J6-M	Brightness	Passive Test Point	Open		Test Point TP 10-M
J6-P	-12 VDC Harmonization	Passive Test Point	Open		Test Point TP 10-P
J6-R	+12 VDC Harmonization	Passive Test Point	Open		Test Point TP 10-R

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Table V - HUD Test Points (Continued)

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J6-S	Chassis Ground	Passive Test Point	Open Fail Test 0 VDC	-0.5 VDC to 0.8 VDC	Test Point TP 10-S To HUD Fail Push Button Switch
J6-T	HUD Fail Test	DC	Normally Open		Test Point TP 10-T
J6-V	Y Yoke Return	Passive Test Point	Open		Test Point TO 10-V
J6-W		Passive Test Point	Open		Test Point TP 10-W
J6-X	Y Sweep Slow	Passive Test Point	Open		Test Point TP 10-X
J6-Y	Y Input Slow	Passive Test Point	Open		Test Point TP 10-Y
J6-Z	Sweep Slow Delay	Passive Test Point	Open		Test Point TP 10-Z
J6-D	Spare	No Connection	Open		Test Point TP 10-D
J6-N	Spare	No Connection	Open		Test Point TP 10-N
J6-U	Spare	No Connection	Open		Test Point TP 10-U
J6-a	Spare	No Connection	Open		Test Point TP 10-a
J6-b	Spare	No Connection	Open		Test Point TO 10-b
J6-c	Spare	No Connection	Open		Test Point TP 10-c

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Table VI - Test Set Outputs to CGSE

Test Set Connector	Signal Name	Signal Type	Signal Level	Signal Limits	Characteristics and/or Remarks
J6-X	X-Deflection	Passive Test Point	Open		Output to Monitor from SP
J6-Y	Y-Deflection	Passive Test Point	Open		Output to Monitor from SP
J6-Z	Bright-Up	Passive Test Point	Open		Output to Monitor from SP

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classed as follows:

a. First article inspection

b. Quality conformance inspection

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three Test Sets selected at random by the Government representative from the first fifteen production Test Sets.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in Table I.

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TABLE I. First article tests

Item	Characteristic	Requirement	Test Procedure
1	Performance	3.2.1	4.6, 4.7.10, 4.7.12
2	Temperature & Humidity	3.2.5.1	4.7.2, 4.7.3
3	Vibration	3.2.5.3	4.7.4
4	Pulse	3.2.5.4.1	4.7.5
5	Transit Drop	3.2.5.4.2	4.7.7
6	Bench Handling	3.2.5.4.3	4.7.6
7	Dripproof	3.2.5.5	MIL-T-28800
8	Salt Atmosphere	3.2.5.6	4.7.8
9	Acoustical Noise	3.2.5.7	MIL-T-28800
10	Fungus	3.2.5.8	4.7.9
11	Electromagnetic Radiation	3.2.5.9	4.7.11, MIL-STD-461
12	Workmanship	3.6	MIL-STD-454
13	Interchangeability	3.3.2	MIL-STD-454
14	Safety	3.3.3	MIL-STD-454
15	Preconditioning	3.3.4	4.7.13

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TABLE II. Performance tests

1	Performance	3.2	4.6
2	Vibration	3.2.5.3	4.7.4
3	Pulse	3.2.5.4.1	4.7.5
4	Transit Drop	3.2.5.4.2	4.7.7
5	Workmanship	3.6	MIL-STD-454
6	Interchangeability	3.3.2	MIL-STD-454
7	Preconditioning	3.3.3	4.7.13

4.5 Quality conformance inspection.

4.5.1 Performance inspection.

4.5.1.1 Sample and tests. Each Test Set in every lot shall be subjected to the tests in Table II and shall be examined visually for completeness, improper assembly, and evidence of poor workmanship.

4.5.1.2 Acceptance. Where any Test Set fails to meet any specified requirement, the defective Test Set shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.1.3 Test equipment. The actual test equipment used during the performance of the test shall be recorded on the applicable data sheets. The nomenclature, type, model, and identification numbers shall be recorded for each equipment used. All measurements shall be made with instruments and test equipment whose accuracy and calibration has been verified periodically to meet MIL-STD-45662 requirements. All instruments and test equipment shall be used in accordance with the manufacturers approved procedures and recommendations.

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4.6 Performance tests. Each Test Set submitted for acceptance shall be subjected to the following tests to determine the functional capabilities.

4.6.1 Resistance checks. Perform the following procedures to verify the Power Supply Load Resistors and the continuity of TEST MONITOR TEST POINTS.

Step 1. Verify the TEST SET POWER, HUD POWER, and SYSTEM/SP POWER switches are in the OFF position and no cables are connected to the Test Set.

Step 2. Verify with the OHM DMM the load resistors in Table VII.

Table VII. Load resistor table

DMM RTN	TEST POINT	OHMS	NON-ADJUSTABLE RESISTORS
E1	E2	2200 ± 22 ohms	R1
E1	E3	27K ± 1.35Kohms	R2
F1	F2	200 ± 2 ohms	R3
F1	F3	200 ± 2 ohms	R4
G1	G2	40 ± 1 ohm	R5
G1	G3	40 ± 1 ohm	R6
F4	E4	110 ± 1.1 ohm	R7
H4	G4	18.1 ± 1 ohm	R8

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4.6.2 Continuity check for SP test points and cable.

a. Connect cables W1, W2, W3, W4, and W5 to appropriate test set connectors.

b. Connect an OHM DMM between the following points and verify continuity.

Test Monitor SP Test Points	SP Test Points W4P2
A	A
B	B
C	C
D	D
E	E
F	F
G	G
H	H
J	J
K	K
L	L
M	M
N	N
P	P
R	R
S	S
T	T
U	U
V	V
W	W
X	X
Y	Y
Z	Z
a	a
b	b
c	c
d	d
e	e
f	f
g	g
h	h
i	i
j	j
k	k

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Continuity check for SP test points and cable

Test Monitor SP Test Points	SP Test Points W4P2
m	m
n	n
p	p
q	q
r	r
s	s
t	t
u	u
v	v
w	w
x	x
y	y
z	z
AA	AA
BB	BB
CC	CC
DD	DD
EE	EE
FF	FF
GG	GG
HH	HH
JJ	JJ
LL	LL
MM	MM
NN	NN
PP	PP

Test Monitor To SP Test Points	To SP W2P2
D-2	C
C-3	D
A-1	F
A-4	N
B-4	P
D-3	R
C-2	X

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Continuity check for SP test points and cable

Test Monitor To SP Test Points	To SP W2P2
B-3	Y
A-3	Z
A-2	g
B-1	i
D-1	j
C-1	k
B-2	q
C-4	FF
D-4	EE

Test Monitor From SP Test Points	From SP W3P1
G-4	B
G-2	D
G-1	E
E-1	X
E-3	Y
E-4	b
F-4	c
H-4	d
G-3	f
E-2	u
F-3	FF
F-2	HH
F-1	NN

4.6.3 Continuity check of the HUD test points and cable.

Test Monitor HUD Test Points	HUD Test Points W5P2
A	A
B	B
C	C
E	E
F	F
G	G
H	H

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Continuity check of the HUD test points and cable

Test Monitor HUD Test Points	HUD Test Points W5P2
J	J
K	K
L	L
M	M
P	P
R	R
V	V
W	W
X	X
Y	Y
Z	Z

Test Monitor HUD Test Points	HUD Test Points W5P2
L-4	B
L-2	D
L-1	E
J-1	X
J-3	Y
J-4	b
K-4	c
M-4	d
L-3	f
J-2	u
K-3	FF
K-2	HH
K-1	NN
J-5	J
J-6	K
K-6	P
K-5	R
L-5	j
L-6	m

4.6.4 Panel light test.

a. Connect W3 to J4 and J1 on Unit Under Test.

b. Measure +28 Vdc with DMM + lead at W3P2-W and - lead at W3P2-s when panel light switch is ON.

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4.6.5 Day/night switch test.

a. Connect W3 to J4 and J7 on Unit Under Test.

b. Measure +28 Vdc with DMM + lead at W3P2-EE and - lead at W3P2-S when switch is in NIGHT position.

4.6.6 Power supplies. Perform the following procedures to verify that the power supplies meet the voltage, current, and ripple requirements in Table II.

a. Connect cable W1 to 28 Vdc facility power.

b. Connect cable W3P1 from J3 (from SP) to J4 (to HUD) and J7 to HUD Test Set Front Panel.

c. Position TEST POWER, HUD POWER, and SYSTEM/SP POWER switches to the ON position.

d. Connect the DMM RTN as specified for each power supply to be checked.

e. Verify the below Power Supply voltages at the specified test points:

Test monitor to HUD test points

DMM RTN	DMM+	Voltages	Power Supply	RMS Ac Ripple
J1	J2	+85 ± 10 Vdc	PS1	0.5V
J1	J3	-85 ± 10 Vdc	PS2	0.5V
K1	K2	+40 ± 1 Vdc	PS3	85mV
K1	K3	-40 ± 1 Vdc	PS4	85mV
L1	L2	+15 ± .15 Vdc	PS5	35mV
L1	L3	-15 ± .15 Vdc	PS5	35mV
K4	J4	+26.5 ± 1 Vdc	PS6	55mV
M4	L4	+10.0 ± 1 Vdc	PS7	50mV

Test monitor test set internal

DMM RTN	DMM+	Voltages	Power Supply	RMS Ac Ripple
TP-1	TP-2	+5 ± .5 Vdc	PS8	15mV

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OUTPUT CONNECTOR	AC DMM RTN	AC DMM	VOLTAGE	POWER SUPPLY
J-2	J2-W	J2-A	115 $\frac{+3}{-7}$ VAC	PS-9

f. Remove the DMM leads.

g. Connect the oscilloscope channel one to J2-A and GND to J2-W.

h. Verify 400 ± 40 Hz.

Table VIII. Power supplies requirements.

PS1	+85 ± 10 Vdc	.5V
PS2	-85 ± 10 Vdc	.5V
PS3	+40 ± 1 Vdc	85mV
PS4	-40 ± 1 Vdc	85mV
PS5	+15 ± .15 Vdc	35mV
PS6	+26.5 ± 1 Vdc	55mV
PS7	10 ± 1 Vdc	50mV
PS8	+5 ± .5 Vdc	15mV
PS9	115 $\frac{+3}{-7}$ VAC	

4.7 Performance test under simulated environmental conditions.

4.7.1 Environmental test requirements and procedures.

4.7.1.1 Test sequence. The Test Set shall have completed the test of 4.6 prior to performance of the detailed environmental tests herein. The performance test of 4.6 shall be performed before, during, or after each environmental test as required for the purpose of determining the performance of the Test Set. The test of 4.6 shall again be performed after all environmental tests have been completed.

4.7.2 Temperature/altitude. The Test Set shall be subjected to the applicable tests required by MIL-STD-810 with the temperature requirements modified by MIL-T-28800. The applicable tests are for Category 2 equipment.

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4.7.2.1 Test setup.

a. The Test Set shall be instrumented with small diameter thermocouples placed on components within the unit, which are expected to exhibit the greatest thermal inertia during the test. All components instrumented with thermocouples shall be identified and listed on the applicable data sheet. The thermocouple leads shall be routed to the outside of the Test Set by any convenient means.

b. The test setup and methods of operation of the Test Set shall be such as to simulate actual usage insofar as practical. The Test Set undergoing tests shall be placed in the test chamber with electrical and thermocouple leads made through altitude sealed terminals, connectors, and parts. The thermocouple leads shall be connected to a multi-point temperature recorder.

Notes: The following guidelines are provided for consideration when determining location of thermal sensors used to monitor the test items:

1. One or more sensor in the ambient air within each major unit.

2. Contact temperature on the largest mass in each major unit.

3. Contact temperature on the part(s) where the highest surface temperature is expected.

4. Contact temperature on the part(s) whose temperature is likely to limit equipment performance.

4.7.3 Humidity. The FCC-FTS shall be subjected to the applicable tests required by MIL-STD-810 as modified by MIL-T-28800.

4.7.3.1 Test facility.

a. Humidity chamber. The humidity tests shall be performed in a chamber capable of generating and controlling relative humidity in excess of 95% RH by steam or evaporation of tap water having pH value between 6.0 and 7.2 measure at 23°C. The chamber shall also contain provisions for limiting air velocity throughout

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the exposure are to 150 feet per minute and provisions made to prevent dripping and collecting water-condensate onto the equipment from overhead. The chamber shall have been designed and constructed with provisions for, 1) venting to atmosphere to preclude vapor pressure buildup and 2) sufficient internal volume so that the Test Set under test will not interfere with the generation and control of the test conditions.

b. Humidity sensor. The relative humidity within the test chamber shall be verified by use of one or more of the following type indicators:

(1) Dew Point indicator (Dry bulb temperature and water content).

(2) Differential Temperature Indicator (Dry bulb temperature - wet bulb temperature).

(3) Separate Dry Bulb and Wet Bulb Indicators (with suitable resolution).

The preferred method is (1). If a wet bulb sensor is used, the air flow past the sensor shall be at least 900 feet per minute. Both the dry bulb and wet bulb sensors shall be located in the same air stream with the dry bulb sensor oriented beside or upstream from the wet bulb sensor.

4.7.3.2 Test setup. The test setup of the Test Set during the humidity cycle shall be such as to simulate actual usage insofar as practical. Plugs, covers, and inspection plates not used in operation, but used in servicing, shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test item is installed. The Test Set shall be oriented within the chamber so that water condensation will not form and collect at any one localized spot.

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4.7.3.3 Post humidity operation. At the completion of the five humidity cycles, the Test Set shall be removed from the humidity chamber and subjected to the post operational performance tests in accordance with 4.6 of this procedure as soon as possible, but not later than one hour after completion of the humidity test. Prior to measurements, excess moisture may be removed from the exterior surfaces of the Test Set by turning the unit upside down or by wiping the external surfaces only.

4.7.3.4 Moisture test. Following the above check for satisfactory operation, the Test Set shall be opened and examined for evidence of entrapped moisture.

4.7.4 Vibration. The Test Set shall be subjected to the tests required by MIL-STD-810 and MIL-T-28800.

4.7.4.1 Test facility.

a. Vibration exciter system. The vibration tests shall be performed on a vibration exciter system capable of shaking the Test Set at the specified amplitudes and frequencies. Transverse action of the exciter (unloaded) shall be less than 20 percent of the input motion and the harmonic wave distortion of the fundamental frequency shall not exceed 10 percent.

Test conditions

1. Resonant Search 5-55 Hz (each of 3 axes) non-operating (2 increments).

2. Resonant Dwell (each of 3 axes) non-operating.

3. Sinusoidal Cycling 5-55 Hz (each of 3 axes) non-operating.

b. Accelerometer, control. The accelerometer used to measure and control the input acceleration to the test specimen shall be rigidly attached to the base plate of the test specimen vibration fixture, using the stud normally supplied or by cementing in place using Eastman 910 or dental adhesives. The accelerometer shall be positioned as close as possible to the attachment points of the Test Set to the fixture. The frequency response of the control accelerometer should be flat within $\pm$ 5%, 5 Hz to 3 KHz at room ambient temperature.

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c. Test fixture. The holding fixture for the Test Set shall simulate the normal installation as nearly as possible. That is it shall have, 1) provisions for attachment of its base plate directly to the vibration exciter head or the horizontal slip table, and 2) provisions for the rigid attachment of the Test Set to the base plate.

4.7.4.2 Test setup.

a. Test set and test station. The test setup of the Test Set shall be such as to simulate equipment transported as secured cargo. The test set cover shall be closed and secure. The Test Set to the vibration exciter (head or slip table as required) using the normal mounting attachment points. The test equipment shall be positioned as far away from the vibration exciter as possible to limit the effect of stray magnetic fields. The limiting factors, however, are the length of the interconnecting cables. The Test Set will not be operated during vibration.

b. Interconnecting cables. All interconnecting cables from the Test Set to the test station shall be suspended by elastic cord (bungee), positioned at a point approximately six inches from the Test Set termination. All cables, including accelerometer, shall be oriented so that they emanate in a straight line away from the center of the vibration exciter.

c. Accelerometers (equipment). The Test Set under test shall be equipped with accelerometers located on the unit's surface in each of the three perpendicular axes for determining resonance frequencies and acceleration "g".

4.7.4.3 Vibration test procedure.

4.7.4.3.1 Resonance search. Resonant frequencies of the equipment shall be determined by varying the frequency of applied vibration slowly through the specified range at reduced test levels but with sufficient amplitude to excite the item. Sinusoidal resonance search may be performed using the test level and cycling time specified for sinusoidal cycling test, provided the resonance search time is included in the required cycling test time.

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4.7.4.3.2 Resonance dwell. The test item shall be vibrated along each axis at the mode severe resonant frequencies determined in the resonance search. Test levels, frequency ranges, and test times shall be in accordance with Table IX. If more than four significant resonant frequencies are found for any one axis, the four mode severe resonant frequencies shall be chosen for the dwell test. If a change in the resonance frequencies occurs during the test, its time of occurrence shall be recorded and immediately the frequency shall be adjusted to maintain the peak resonance condition.

Table IX. Resonance dwell

Frequency Increments (Hz)	Vibratory Level (Inches Peak-to-Peak)	Sweep Time per increment* (minutes)	Survey Time per Axis (minutes)	Resonance Dwell (minutes)
5 to 25	0.025 inches (0.635mm)	10	15	10
25 to 55	0.020 inches (0.508mm)	5		

*Note: Sweep times specified shall be for a low frequency to high frequency to lot frequency variation for the applicable increment.

4.7.4.3.3 Sinusoidal cycling. The test item shall be vibrated thru one cycle along each axis in accordance with the displacement levels, frequency range (5-55 Hz) and time (15 min.). The accumulated minimum vibration test time per axis shall be 45 minutes. The specified sweep time (15 min.) is that of an ascending sweep only.

4.7.4.3.4 Post-vibration operation. At the completion of vibration testing, the post operational tests of 4.6 shall be performed.

4.7.5 Shock. The Test Set unit shall be subjected to the applicable tests required by MIL-STD-810 and MIL-T-28800.

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4.7.5.1 Test conditions.

Equipment shock

2 directions in each of 3 axes
(3 shocks each, 15g, half sine)
(11 msec non-operating)
(Total of 18 shocks)

4.7.5.2 Test facility.

a. The shock tests shall be performed on a shock generating system capable of exciting the test specimen to the required waveform, pulse amplitude, and time duration. The overall system response ratio, output divided by input (Eo/Ei), shall be within the limits specified below:

Frequency range	Response (Eo/Ei)
2 to 4 Hz	1.0 ± 0.1, -0.3 @ 2 Hz; -0.1 4 Hz
4 to 2000 Hz	1.0 ± 0.1
2 KHz to 7 KHz	1.0 ± 0.1 @ 2 KHz; -1.0 7 KHz

b. Accelerometer. The accelerometer used to measure the input shock to the Test Set shall be rigidly attached to the baseplate of the Test Set shock fixture, using the stud normally supplied or by cementing in place using Eastman 910 or dental adhesives. The accelerometer shall be positioned as close as possible to the attachment points of the Test Set to the fixture. The accelerometer resonant frequency shall be greater than 14 KHz (30 KHz or higher is recommended) for piezo-electric types. For suitable low frequency response, the accelerometer and load (cathode - follower, amplifier, or other) shall have the following characteristics:

RC 0.02 Hz

Where

R - Load Resistance (ohms)

C - Transducer Capacitance plus shunt capacitance of cable and load (farads)

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Strain-gage type transducers may be used provided the undamped natural frequency is equal or greater than 1.5 KHz with the damping approximately 0.64 to 0.70 of critical. (Resonant frequencies greater than 1.5 KHz are recommended.)

c. Filters. A band-pass filter may be used to display and record the fundamental frequency of the shock pulse providing it has band-pass within the frequency range of 0.2 to 650 Hz and will not degrade the response of the system to exceed the limits specified in item a, above.

d. Fixture. The holding fixture for the Test Set shall be the same test fixture used for the vibration test.

4.7.5.3 Test setup. The test setup shall simulate handling, transportation, and service use. The Test Set undergoing test and the shock fixture shall be securely attached to the shock machine carriage using the normal mounting attachment points.

4.7.5.4 Equipment shock. With the Test Set non-operating, it shall be subjected to three shocks in each of two places in each of the three mutually perpendicular axes. The sequence of test may be in any order.

a. Z' plane

b. Z plane

c. Y' plane

d. Y plane

e. X' plane

f. X plane

The shock pulse characteristics are summarized as follows:

Waveform	-	Half sine
Amplitude	-	15 g $\pm$ 3.0 g
Time Duration	-	11 $\pm$ 0.77 milliseconds

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4.7.5.5 Requirements. The shock input spectrum (amplitude, period, and waveform) shall be monitored for each shock, however only one photograph of the shock spectrum need be made to assure proper system performance and calibration. During the shock test there is no requirement for functional performance. After each series of shocks in each plane, the Test Set mounting attachment points shall be examined for evidence of mechanical failure. After the completion of all 18 shocks (3 axes x 2 planes x 3 shocks each) the Test Set shall be carefully examined for evidence of mechanical failure with all covers open.

4.7.5.6 Post shock operation. At completion of test, the Test Set shall be operated per 4.6 and inspected.

4.7.6 Bench handling. The Test Set shall be subjected to the applicable tests required by MIL-STD-810 and MIL-T-28800.

4.7.6.1 Bench handling test. Bench handling tests should be performed on the Test Set as follows:

The chassis and cover assemblies shall be removed from its enclosure, as for servicing and placed in a suitable position for servicing on a horizontal, solid, wooden bench top at least 1-5/8 inches thick. Using one edge as a pivot, lift the opposite edge of the chassis until one of the following conditions occur (whichever occurs first):

a. The chassis forms an angle of 45 degrees with the horizontal bench top.

b. The lifted edge of the chassis has been raised 4 inches above the horizontal bench top.

c. The lifted edge of the chassis is just below the point of perfect balance.

Let the chassis drop back freely to the horizontal bench top. Repeat, using other practical edges of the same horizontal face as pivot points for a total of four drops.

Repeat the above procedure with the test item resting on other practical edges until it has been dropped for a total of four times on each face which the test item could be placed practically during servicing.

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4.7.6.2 Post bench handling inspection/operation. At the conclusion of the test, the Test Set shall be carefully examined for evidence of mechanical failure. It shall then be reassembled, connected, and operated in accordance with the post operational tests of 4.6.

4.7.7 Transit drop. The Test Set shall be subjected to the applicable tests required by MIL-T-28800.

4.7.7.1 Transit drop test. The transit drop test shall be performed on the Test Set as follows:

a. The combination case including a full compliment of equipment (i.e., panel assembly, logic assembly, cables, manual) shall be placed in the case and the cover shall be locked in place.

b. The Test Set shall be dropped once on each corner using a quick-release hook from a height of 12 inches onto 2-inch plywood backed concrete (8 total).

c. Repeat for each edge (12 total).

d. Repeat for each face (6 total).

For a total of 26 drops.

4.7.7.2 Post drop inspection/operation. Upon completion of the transit drops, the Test Set shall be operated per 4.6 and inspected. There shall be no evidence of damage to the case, such as openings along any joint, loosening of partitions, splitting or cracking of case material. Any partial or total opening of catches shall be cause for rejection of the case. The case and cover shall function in the same manner as prior to the drop tests.

4.7.8 Salt atmosphere. The Test Set shall be subjected to MIL-STD-810. Operation of the Test Set shall not be required during testing and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish of metal surface.

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4.7.9 Fungus. The Test Set shall be subjected to the test of MIL-STD-810. Gasket sealed or other non-hermetic sealed test items shall be opened during test exposure and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required during testing, but examination of the concentration after exposure period shall reveal no evidence of viable fungus on any surface.

4.7.10 Explosive atmosphere. The Test Set shall be subjected to MIL-STD-810, except test shall be limited to a maximum altitude of 20,000 feet.

4.7.11 Electromagnetic radiation. The Test Set shall be installed in laboratory test fixtures in a manner that will simulate actual service usage and tested to MIL-STD-461.

4.7.12 Sand and dust. The Test Set shall be subjected to tests of MIL-STD-810, except that the air velocity shall be 200 ± 100 FPM.

4.7.13 Preconditioning (burn-in). Subject each Test Set to burn-in.

4.7.13.1 Burn-in requirements.

4.7.13.1.1 Burn-in. Burn-in is to be applied to all deliverable systems.

4.7.13.1.2 Test environments. Environmental exposure for the Test Set during burn-in is Level F of MIL-STD-781. Cycle time shall be determined by thermal survey. Chamber temperature change shall be programmed equal to or greater than 5° per minute. One cycle of the environmental profile is described below.

Temperature	-54°C to +71°C (-65°F to 160°F).
Temperature Cycling	Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus 2 hours.
Vibration	2.2G ± 10% peak acceleration value at 30 ± 5 Hz measured at the mounting points. The duration of vibration shall be at least 10 minutes during each hour of equipment operation time.

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Equipment On-Off Cycling	Equipment off during cooling cycle and on during heating cycle.
Input Voltage	Specified voltage +5 -2%.
Input Voltage Cycling	a. Nominal voltage for first 1/2 of "ON" time starting at the beginning of the heating period. b. Momentary off. c. 100% of nominal for each 1/3 of "ON" time. d. Momentary off. e. 90% of nominal for next 1/3 of "ON" time.

4.7.13.1.3 Number of cycles.

4.7.13.3.1 Minimum number. The minimum number of cycles shall be five (5) with the last two (2) cycles failure free.

5. PACKAGING.

5.1 Preservation, packing, packaging, and marking. The Test Set shall be preserved, packaged, packed, and marked in accordance with the requirements of the preservation and packaging data sheet at the level of protection specified in the contract.

5.2 Commercial packaging. Commercial packaging shall be in accordance with ASTM-D3951.

6. NOTES.

6.1 Intended use. This equipment is intended for the functional test and fault isolation of the Head-Up Display, M76.

6.2 Ordering data.

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6.2.1 Acquisition requirement. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable Packaging Data Sheet.

c. Selection for applicable levels of preservation, packaging, and packing.

d. Applicable stock number.

e. First article inspection requirements.

6.3 Subject term (key word) listing.

Circuit Card
Display Unit
Extender Board
Electrical Interface
Fault Isolation
Head Up Subsystem
Head Up Display
Microcircuits
Signal Processor
Test Points
Test Set

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1270-A058)

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